

Question number	Answer	Additional guidance	Mark
3(a)	An answer that includes the following points: Similarity: - both target bacteria (1) Differences: {bacteriostatic antibiotics / tetracycline} prevent {cell division / growth} and {bactericidal / vancomycin} {kill / destroy} (bacteria) (1)	ACCEPT both affect bacteria both used to treat bacterial infections both bacteria even if qualification is wrong - this would negate mp 2 not mp 1 IGNORE description of mechanisms named antibiotics	(2)

Question number	Answer	Additional guidance	Mark
3(b)(i)	A description that includes three of the following points: - antibiotics dissolved in a solvent (1) (then) antibiotics loaded onto the {filter paper / origin} (1) - filter paper (then) dipped into solvent (below the origin) (1) - and (then) left to run until the leading edge is near end of filter paper) (1)	ACCEPT named solvent e.g. water, ethanol ACCEPT added to the filter paper / chromatogram / TLC plate / other appropriate medium IGNORE onto glass ACCEPT named solvent	(3)

Question number	Answer	Mark
3(b)(ii)	The only correct answer is B <i>A is incorrect because the tetracycline spot is about three-quarters of the way up so Rf value is approx. 0.75</i> <i>C is incorrect because the tetracycline spot is about three-quarters of the way up so Rf value is approx. 0.75</i> <i>D is incorrect because the tetracycline spot is about three-quarters of the way up so Rf value is approx. 0.75</i>	(1)

Question number	Answer	Additional guidance	Mark
3(b)(iii)	An explanation that includes two of the following points: - because different solvents were used (1) - difference in solubilities (1) - tetracycline is more soluble (in both solvents) / vancomycin is less soluble (in both solvents) / both antibiotics are more soluble in the solvent used in method 2 (1) - different solid phase used (1)	ACCEPT two named solvents ACCEPT the more soluble the further it will travel antibiotics bind to medium differently different charge / molecular mass IGNORE dissolve differently IGNORE size / mass / structure / properties ACCEPT converse for method 1	(2)

